

Feedback Control Prelab 2 DSW97410

$$\textcircled{1}. H(s) \Big|_{s=j\omega} = \frac{1}{\left(\frac{j\omega}{\omega_c}\right)^3 + 2\left(\frac{j\omega}{\omega_c}\right)^2 + 2\left(\frac{j\omega}{\omega_c}\right) + 1}$$

$$H(j\omega) = \frac{1}{-j\left(\frac{\omega}{\omega_c}\right)^3 - 2\left(\frac{\omega}{\omega_c}\right)^2 + j2\left(\frac{\omega}{\omega_c}\right) + 1}$$

$$\text{let } \frac{\omega}{\omega_c} = \gamma$$

$$H(j\omega) = \frac{1}{-j\gamma^3 - 2\gamma^2 + j2\gamma + 1}$$

$$= \frac{1}{1 - 2\gamma^2 + j(2\gamma - \gamma^3)}$$

$$|H(j\omega)| = \frac{1}{\sqrt{(1 - 2\gamma^2)^2 + (2\gamma - \gamma^3)^2}}$$

$$1 - \cancel{4\gamma^2} + \cancel{4\gamma^4} + \cancel{4\gamma^2} - \cancel{4\gamma^4} + \gamma^6 = 1 + \gamma^6$$

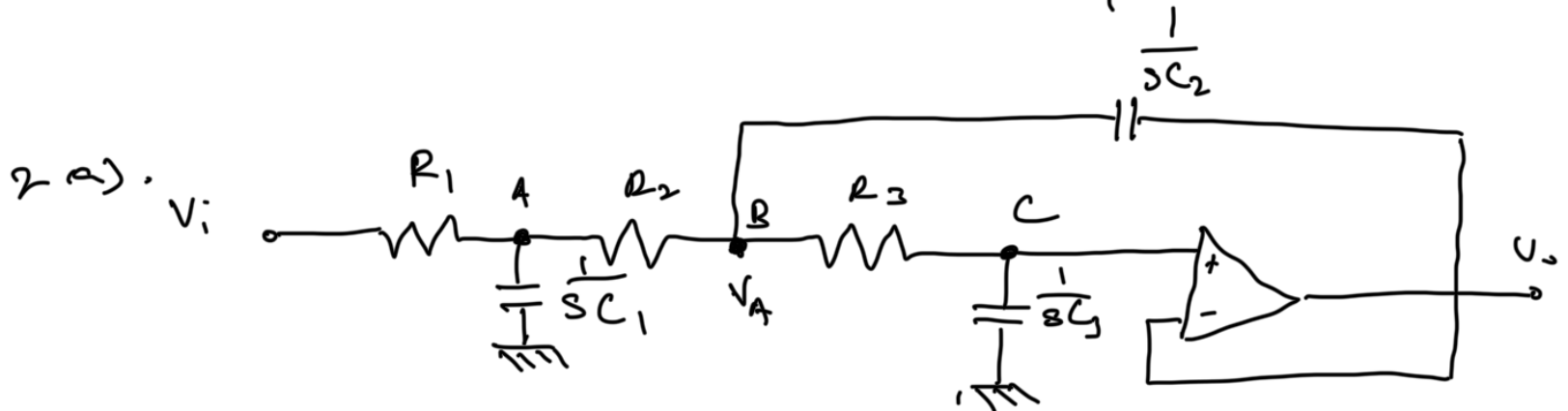
$$\therefore |H(j\omega)| = \frac{1}{\sqrt{1 + \gamma^6}} \Big|_{\gamma = \frac{\omega}{\omega_c}}$$

$$= \frac{1}{\sqrt{1 + \left|\frac{\omega}{\omega_c}\right|^6}}$$

$$\text{From 1.1 } |H(j\omega)| = \underline{\underline{\frac{1}{\sqrt{1 + \left|\frac{\omega}{\omega_c}\right|^6}}}}$$

$$\sqrt{1 + \left| \frac{\omega}{\omega_c} \right|^{2n}} \Big|_{n=3}$$

$$= \frac{1}{\sqrt{1 + \left| \frac{\omega}{\omega_c} \right|^6}}$$



@C Using nodal analysis

$$V_c = V_o$$

$$\frac{-V_c + V_B}{R_3} = \frac{V_c}{\frac{1}{sC_3}}$$

$$-V_o + V_B = V_o s R_3 C_3$$

$$\therefore V_B = V_o (1 + s R_3 C_3)$$

@B

$$\frac{V_B - V_A}{R_2} + \frac{V_B - V_o}{R_3} + (V_B - V_o) s C_2 = 0$$

$$\frac{V_o (1 + s R_3 C_3) - V_A}{R_2} + \frac{V_o (1 + s R_3 C_3) - V_o}{R_3} + (V_o (1 + s R_3 C_3) - V_o) s C_2 = 0$$

$$V_A = V_o \left[1 + s(R_3 C_3 + R_2 C_3 + R_2 C_2) + s^2 (R_2 R_3 C_2 C_3) \right]$$

@A

$$\frac{V_A - V_B}{R_2} + \frac{V_A - V_i}{R_1} + V_A s C_1 = 0$$

$$V_i = V_A \left(1 + \frac{R_1}{R_2} + s R_1 C_1 \right) - V_B \left(\frac{R_1}{R_2} \right)$$

Substituting V_A & V_B

$$\therefore \frac{V_o(s)}{V_i(s)} = \frac{1}{a_3 s^3 + a_2 s^2 + a_1 s + 1}$$

Where $a_3 = R_1 R_2 R_3 C_1 C_2 C_3$

$$a_2 = R_3 C_2 C_3 (R_1 + R_2) + C_1 C_3 R_1 (R_2 + R_3)$$

$$a_1 = R_1 C_1 + C_3 (R_1 + R_2 + R_3)$$

(b), $\bar{C}_1 = 1.392$, $\bar{C}_2 = 3.846$, $\bar{C}_3 = 0.2024$

$$f_c = 50 \text{ Hz}$$

$$R = 44.3 \text{ k}\Omega$$

$$\omega_c = 2\pi f_c = 2\pi (\text{sec}) = 314.16 \text{ rad/sec}$$

$$C'_i = \frac{\bar{C}_i}{\omega_c} \quad C_i = \frac{C'_i}{R_1}$$

$$\therefore C_i = \frac{\bar{C}_i}{R \cdot \omega_c}$$

$$C_1 = \frac{1.392}{44.3 \times 10^3 \times 314.16} \approx 100 \text{ nF}$$

$$C_2 = \frac{3.846}{44.3 \times 10^3 \times 314.16} \approx 288 \text{ nF}$$

$$C_3 = \frac{0.2024}{44.3 \times 10^3 \times 314.16} \approx 14.8 \text{ nF}$$

$$H(s) = \frac{1}{a_3 s^3 + a_2 s^2 + a_1 s + 1}$$

$$\begin{aligned}
 a_3 &= (44.3 \times 10^3)^3 \times 100 \times 255 \times 14.3 \times 10^{-17} \\
 &= 3.2143 \times 10^{-8} \\
 &\approx (314.5)^{-3}
 \end{aligned}$$

$$\begin{aligned}
 a_2 &= 2(44.3 \times 10^3)^2 \times 255 \times 14.3 \times 10^{-16} + 100 \times 255 \times 10^{-18} \times 2 \\
 &\quad \times (44.3 \times 10^3)^2 \\
 &\approx (314.5)^{-2}
 \end{aligned}$$

$$\begin{aligned}
 a_1 &= 100 \times 10^{-9} \times 44.3 \times 10^3 + 14.3 \times 10^{-9} \times 3(44.3 \times 10^3) \\
 &\approx 314.5^{-1}
 \end{aligned}$$

$$\therefore H(s) = \frac{1}{\left(\frac{s}{314.5}\right)^3 + \left(\frac{s}{314.5}\right)^2 + \left(\frac{s}{314.5}\right) + 1}$$

which matches the transfer function

Q3

$$f_s = 1 \text{ kHz} \quad x(n) = \sin(0.1\pi n)$$

$$x(n) = x'(nT_s)$$

$$T_s = \frac{1}{f_s} = \frac{1}{1 \text{ kHz}} = 1 \text{ ms}$$

$$x(n) = \sin(\omega n) \Rightarrow \omega = 0.1\pi$$

$$\omega = 2\pi f$$

$$f = \frac{\omega f_s}{2\pi} = \frac{0.1\pi \times 1000}{2\pi} = 50 \text{ Hz}$$

$$\omega_f = 2\pi f = 2\pi \times 50 = 100\pi$$

$$\therefore x(t) = \sin(100\pi t)$$

$$b) \sin(\omega n) = \sin((\omega + 2\pi k)n) \quad \forall k \in \mathbb{Z}$$

$$\omega f_s = 2\pi f_k \Rightarrow \omega = 2\pi \frac{f_k}{f_s}$$

$$\therefore x(n) = \sin\left(2\pi \frac{f_k}{f_s} n\right)$$

$$\frac{2\pi f_k}{f_s} = 0.1\pi + 2\pi k$$

$$\therefore \omega_{f_k} = (0.1\pi + 2\pi k) 10^3$$

$$\therefore \omega_{f_k} = (100\pi + 2\pi \times 10^3 k)$$

$\therefore$ All possible Analog waveforms

$$x(t) = \sin((100\pi + 2\pi \times 10^3 k)t) \quad \forall k \in \mathbb{Z}$$

Assuming no aliasing, $k=0$

$\therefore$ part A result is obtained.